Part II. Retrieval

> *How do we get all the data in the right place to train a recommendation system? How do we build and deploy systems for real-time inference?*

Reading research papers about recommendation systems will often give the impression that they're built via a bunch of math equations, and all the really hard work of using recommendation systems is in connecting these equations to the features of your problem. More realistically, the first several steps of building a production recommendation system fall under systems engineering. Understanding how your data will make it into your system, be manipulated into the correct structure, and then be available in each of the relevant steps of the training flow often constitutes the bulk of the initial recommendation system's work. But even beyond this initial phase, ensuring that all the necessary components are fast enough and robust enough for production environments requires yet another significant investment in platform infrastructure.

Often you'll build a component responsible for processing the various types of data and storing them in a convenient format. Next, you'll construct a model that takes that data and encodes it in a latent space or other representation model. Finally, you'll need to transform an input request into the representation as a query in this space. These steps usually take the form of jobs in a workflow management platform or services deployed as endpoints. The next few chapters will walk you through the relevant technologies and concepts necessary to build and deploy these systems—and the awareness of important aspects of reliability, scalability, and efficiency.

You might be thinking, "I'm a data scientist! I don't need to know all this!" But you should know that RecSys has an inconvenient duality: model architecture changes often affect the systems architecture. Interested in trying out those fancy transformers? Your deployment strategy is going to need a new design. Maybe your clever feature embeddings can solve the cold-start problem! Those feature embeddings will need to serve your encoding layers and integrate with your new NoSQL feature store. Don't panic! This part of the book is a walk through the Big Data Zoo.